

LUHUAN WU

Website: luhuanwu.github.io
Email: lwu@flatironinstitute.org
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ACADEMIC POSITION

Flatiron Institute
Associate Research Scientist

New York, NY
2025 - Current

EDUCATION

Columbia University
Ph.D. in Statistics. Advisors: John P. Cunningham and David M. Blei
M.Phil. in Statistics
M.S. in Data Science

New York, NY
2020 - 2025
2020 - 2024
2018 - 2020

Nanjing University
B.S. in Mathematics

Nanjing, China
2014 - 2018

RESEARCH INTERESTS

- Generative models
- Approximate inference and sampling
- Probabilistic modeling and Bayesian methods

PUBLICATIONS & PREPRINTS

* for equal contribution

1. Reverse Diffusion Sequential Monte Carlo. *NeurIPS 2025*
Luhuan Wu, Han Yi, Christian A. Naesseth, John P. Cunningham
2. Bayesian Modeling of Data from Heterogeneous Environments. *arXiv:2506.22675*
Luhuan Wu, Mingzhang Yin, Yixin Wang, John P. Cunningham, and David M. Blei
3. Posterior Uncertainty Quantification in Neural Networks using Data Augmentation. *AISTATS 2024*
Luhuan Wu, Sinead Williamson
4. Practical and Asymptotically Exact Conditional Sampling in Diffusion Models. *NeurIPS 2023*
Luhuan Wu*, Brian L. Trippe*, John P. Cunningham, and David M. Blei
5. Denoising Deep Generative Models. *NeurIPS 2022 I Can't Believe It's Not Better Workshop*
Gabriel Loaiza-Ganem, Brendan Leigh Ross, **Luhuan Wu**, John P. Cunningham, Jesse C. Cresswell, Anthony L. Caterini
6. Variational Nearest Neighbor Gaussian Processes. *ICML 2022*
Luhuan Wu, Geoff Pleiss, and John P. Cunningham
7. Bias-free Scalable Gaussian Processes via Randomized Truncations. *ICML 2021*
Andres Potapczynski*, **Luhuan Wu***, Dan Biderman*, Geoff Pleiss, and John P. Cunningham

8. Hierarchical Inducing Point Gaussian Process for Inter-domain Observations. *AISTATS 2021*
Luhuan Wu*, Andrew Miller*, Lauren Anderson, Geoff Pleiss, David M. Blei, and John P. Cunningham
9. Inverse Articulated-body Dynamics from Video via Variational Sequential Monte Carlo.
NeurIPS 2020 Workshop on Differentiable Vision, Graphics, and Physics in Machine Learning
Dan Biderman, Christian A. Naesseth, **Luhuan Wu**, Taiga Abe, Alice C. Mosberger, Leslie J. Sibener, Rui Costa, James Murray, John P. Cunningham
10. Variational Objectives for Markovian Dynamics with Backward Simulation. *ECAI 2020*
Antonio Khalil Moretti*, Zizhao Wang*, **Luhuan Wu***, Iddo Drori, Itsik Pe'er

PROFESSIONAL EXPERIENCE

Apple, AIML

Research Intern

Seattle, WA

May 2023 - Sept 2023

Conduct research on statistical uncertainty quantification methods for deep learning models.
Develop a method based on Bayesian nonparametrics and data augmentation techniques.

Columbia University, Zuckerman Institute

Research Staff

New York, NY

Feb 2020 - Sept 2020

Conduct research on statistical machine learning methods for neuroscience applications.
Develop a state space model for analyzing the animal behavioral datasets.

ACADEMIC SERVICE

Reviewer for JMLR (2021,2023,2025), ICML (2023,2024), NeurIPS (2022,2023,2025), ICLR (2024,2025), AISTATS (2022,2023,2024), UAI (2024,2025), AAAI (2025), TMLR (2025), Statistics and Computing (2025)

TALKS

Oct 2025: Department of Statistics, Rutgers University. *New Brunswick, NJ*

Aug 2025: Microsoft Research New England Generative Modeling and Sampling Seminar.
Boston, MA

Jun 2025: The Gatsby Tri-center Annular Meeting. *London, UK*

Apr 2025: The 7th Advances in Approximate Bayesian Inference. *Singapore*

Jun 2024: The 33rd ICSA Applied Statistics Symposium. *Nashville, TN*

Apr 2023: Berkeley–Columbia Meeting in Engineering and Statistics. *New York, NY*

May 2022: The 35th New England Statistical Symposium. *Storrs, CT*